



Contents lists available at ScienceDirect

Journal of Experimental Child Psychology

journal homepage: www.elsevier.com/locate/jecp



Why do children punish? Fair outcomes matter more than intent in children's second- and third-party punishment



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ARTICLE INFO

Article history:

Received 18 September 2019

Revised 13 January 2020

Available online 28 August 2020

Keywords:

Second-party punishment

Third-party punishment

Development

Fairness

Inequity

Intention

ABSTRACT

Humans punish fairness violations both as victims and as impartial third parties, which can maintain cooperative behavior. However, it is unknown whether similar motivations underlie punishment of unfairness in these two contexts. Here we approached this question by focusing on how both types of punishment develop in children, asking the question: What motivates young children to punish in response to fairness norm violations? We explored two potential factors: the direct experience of unfair outcomes and a partner's fair versus unfair intentions. The participants, 5- and 7-year-olds, were given the chance to engage in both second- and third-party punishment in response to either intended or unintended fairness norm violations in a single paradigm. In both age-groups, children were more likely to punish when they were directly affected by the allocation (second-party punishment) than when they were an uninvolved third party (third-party punishment). Reliable third-party punishment was shown only in the older age-group. Moreover, children's punishment was driven by outcome rather than intent, with equal rates of punishment when unequal outcomes were either the result of chance or the intentional act of another child. These findings suggest that younger children may be mainly motivated to create equal outcomes between themselves and others, whereas older children are motivated to enforce fairness norms as a general principle.

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Introduction

Fairness is central to the emergence and maintenance of human cooperative behavior, especially with unknown others. Punishment can maintain adherence to fairness norms (Fehr & Gächter, 2002), and people are often willing to pay a cost to avoid inequity (Fehr & Fischbacher, 2004; Fehr & Gächter, 2003). Experimental studies operationalize costly punishment as a participant paying a cost to levy a cost on another individual (Clutton-Brock & Parker, 1995; Raihani & McAuliffe, 2012; Raihani, Thornton, & Bshary, 2012). Using this approach, a large body of work has shown that adults will pay a cost to punish when they themselves receive less than others (Güth, Schmittberger, & Schwarze, 1982; Thaler, 1988) and will also pay a cost to punish those who are unfair to others even when they are in the role of an unaffected third party (Fehr & Fischbacher, 2004; Henrich et al., 2006, 2010; Mathew & Boyd, 2011; Raihani et al., 2012).

During recent years, studies have begun to adapt fairness tests from adult research to gain insight into the origins of human fairness in children. Specifically, whereas infants already differentiate between unequal and equal sharing events (Sloane, Baillargeon, & Premack, 2012), children's own sharing behavior is only gradually governed by fairness principles—such as equality and merit—during middle childhood and school age (Blake, McAuliffe, & Warneken, 2014; Gummerum, Hanoch, & Keller, 2008; Warneken, 2018). Here we focused on two types of behaviors that have been discussed as being particularly informative regarding the human sense of fairness: inequity aversion and third-party punishment (McAuliffe, Blake, Steinbeis, & Warneken, 2017). Inequity aversion is the idea that one may be averse to unequal distributions of resources. Indeed, experiments with samples from the United States and Europe show that children will pay a cost to prevent unequal distributions that disadvantage the self (Blake et al., 2015) and will reject unfair offers at a cost to the self (Bueno-Guerra, Leiva, Colell, & Call, 2016; Castelli, Massaro, Sanfey, & Marchetti, 2014; Gummerum & Chu, 2014; Güroğlu, van den Bos, & Crone, 2009; Sutter, 2007; Wittig, Jensen, & Tomasello, 2013). Tests of third-party punishment show that children will also pay to punish as a disinterested third party when they observe somebody else being treated unfairly (Jordan, McAuliffe, & Warneken, 2014; McAuliffe, Jordan, & Warneken, 2015; Yang, Choi, Misch, Yang, & Dunham, 2018). These findings have been taken as evidence that a strong sense of fairness emerges around school age, with children applying an equality principle to both themselves and others when valuable resources are at stake (McAuliffe et al., 2017; Shaw & Olson, 2012).

Despite the fact that all punishments in response to fairness norm violations, whether one is the victim or not, seek to produce similar outcomes (correcting a wrong, restoring equality, or deterring future inequality), we propose that the motivations underlying second- and third-party punishment may be different. Whereas punishing as a victim carries immediate direct personal deterrent value (Clutton-Brock & Parker, 1995; Raihani et al., 2012; Trivers, 1971), the same is not true for punishing as an impartial observer. Various theories accord third-party punishment special motivations. Group norm maintenance theories suggest that third-party punishment seeks to maintain norms for the future benefit of the group in general (Boyd, Gintis, Bowles, & Richerson, 2003; Fehr & Fischbacher, 2004; Gintis, Bowles, Boyd, & Fehr, 2003). Social exchange theories suggest that third-party punishment serves to signal to group members that the punisher would make a good cooperative partner in future interactions (Jordan, Hoffman, Bloom, & Rand, 2016; Krasnow, Cosmides, Pedersen, & Tooby, 2012). Although these theories differ in many respects, both propose that punishing as a third party may be relatively more other oriented (concerned with preventing future harm or unfairness to others or with signaling the punisher's cooperative value to group members) when compared with punishing as the victim (concerned with preventing more harm to the self).

To further investigate the relationship between second- and third-party punishment, we sought to answer two questions. First, when do different forms of costly punishment behavior emerge during early childhood? Specifying the relative ontogeny and developmental trajectories of second- and third-party punishment behavior in response to unfairness would provide a clearer understanding of whether these two types of punishment share motivating factors or are driven by distinct concerns. However, no single study has allowed children aged 4 to 7 years—the critical age range when consis-

tent engagement in both of these behaviors emerges—to make costly punishment decisions in *both* second- and third-party contexts, leaving the relative age of emergence of these two types of behaviors ambiguous.

Second, how do children in this critical age range incorporate intent into their second- and third-party punishment judgments? Adults consider both the severity of an outcome *and* the intent driving an agent's behavior when making moral judgments and ascribing punishment (Berg-Cross, 1975; Cushman, 2008; Cushman, Dreber, Wang, & Costa, 2009; Gino, Shu, & Bazerman, 2010; Martin & Cushman, 2015). Do young children use information about intentionality when punishing in the same manner as adults? Research on children's moral evaluations and hypothetical punishment judgments in response to accidental versus intentional personal harm or property damage, suggests that they may (Baird & Astington, 2004; Cushman, Sheketoff, Wharton, & Carey, 2013; Darby & Schlenker, 1989; Helwig, Zelazo, & Wilson, 2001; Nobes, Panagiotaki, & Bartholomew, 2016; Zelazo, Helwig, & Lau, 1996). However, research on children's responses to intended versus unintended unfairness provides a less clear picture. Recently, Engelmann and Tomasello (2019) proposed that young children are strongly motivated to punish in response to intended unfairness because they feel disrespected when treated unfairly. Yet prior research on children's actual punishment behavior in response to unfairness suggests that they do not seem to be taking intent into account (e.g., Sutter, 2007). Here we addressed this paradox by providing a clear and direct test of the role of intent in young children's second- and third-party punishment judgments.

We now address both of these critical questions in detail.

Second-party versus third-party punishment in children

Identifying the age at which children begin engaging in costly third-party punishment relative to second-party punishment would allow us to better specify whether these two types of punishment behaviors share motivating factors or are driven by distinct concerns. If children's concern for fairness norms were the sole motivator of both second- and third-party punishment, both behaviors would be more likely to emerge at approximately the same point in development. However, research in adults and children finds that in cases where individuals are the direct victims of unfairness, their decisions to punish are primarily motivated by a strong emotional response to being treated unfairly rather than purely normative concerns (LoBue, Nishida, Chiong, DeLoache, & Haidt, 2011; McAuliffe, Blake, & Warneken, 2014; Pillutla & Murnighan, 1996; Steinbeis & Singer, 2013). These findings suggest that negative emotional responses to personal experiences of being treated unfairly may motivate subsequent second-party punishment above and beyond concern for enforcing fairness norms.

The role of affective motivators, particularly in response to direct unfair treatment, may facilitate the early emergence of second-party punishment behavior (McAuliffe et al., 2017). In fact, initial evidence suggests that children may begin engaging in costly second-party punishment at an earlier age than costly third-party punishment. Children as young as 4 or 5 years will reject unequal windfall allocations that disadvantage them (Blake & McAuliffe, 2011), and 6- or 7-year-olds will reject unfair offers made by another child in the ultimatum game (Bereby-Meyer & Fiks, 2013). As children get older, they increasingly reject disadvantageous allocations, so that by 7 or 8 years of age they do so almost always (Bereby-Meyer & Fiks, 2013; Blake & McAuliffe, 2011; Bueno-Guerra et al., 2016). Importantly, at 8 years of age children also begin rejecting advantageous allocations (those in which they receive more than the other child; Blake & McAuliffe, 2011), indicating a shift from pure inequity aversion to an increased concern for enforcing fairness norms as they age (McAuliffe et al., 2017).

Young children will also engage in third-party punishment that does not incur a direct cost (Hamlin, Wynn, Bloom, & Mahajan, 2011; Heyman, Loke, & Lee, 2016; Rakoczy, Kaufmann, & Lohse, 2016; Riedl, Jensen, Call, & Tomasello, 2015; Rossano, Rakoczy, & Tomasello, 2011; Vaish, Missana, & Tomasello, 2011). However, punishment in the adult and animal literature is presumed to involve some direct cost to the punisher (Clutton-Brock & Parker, 1995; Raihani et al., 2012). Several studies have found that children do not begin engaging in *costly* third-party punishment in response to fairness norm violations until 6 years of age (Jordan et al., 2014; McAuliffe et al., 2015). Costly third-party punishment rates seem to increase as children age, with 8-year-olds beginning to punish at adult-like rates (Gummerum & Chu, 2014).

Although existing evidence indicates that costly second-party punishment in response to fairness norm violations appears to emerge before costly third-party punishment, these data were gathered from distinct studies using different paradigms and samples, often including norm violations other than fairness. Therefore, drawing clear conclusions about the relative emergence of these two types of punishment behavior is difficult. Only one study has examined both second- and third-party punishment behaviors using the same paradigm in a single set of children. Gummerum and Chu (2014) found that 8-year-old children do engage in both second- and third-party punishment at comparable rates, but they did not test children at the critical age of onset of these behaviors, which is likely 4 to 7 years. Currently, no single study has compared costly second- and third-party punishment rates in children in this critical age range.

The current study addressed this gap in the existing literature by having children aged 4.5 to 7.5 years play both second- and third-party punishment games. Using a within-participant design, children decided whether or not to pay a cost to punish another child who divided candy unequally between herself or himself and the participant (second-party condition) or a third child (third-party condition). In this way, we were able to assess whether costly second-party punishment and costly third-party punishment coemerge during development or whether one type of punishment emerges before the other.

The roles of intent versus outcome in children's punishment behavior

A second factor that may motivate people to punish is whether or not a fairness norm was violated intentionally. Adults' judgments reflect a strong influence of intentionality both when making second-party punishment judgments (Blount, 1995; Falk, Fehr, & Fischbacher, 2008; Nelson, 2002) and when making third-party punishment judgments (Buckholtz et al., 2008; Gummerum & Chu, 2014; Treadway et al., 2014). Only a few studies have looked at the role of intent in children's decision to punish in response to fairness norm violations. In general, these studies suggest that children under 12 years of age do not take intent into account when making punishment decisions in response to fairness norm violations (Bueno-Guerra et al., 2016; Gummerum & Chu, 2014; Güroğlu et al., 2009; Sutter, 2007). These findings are in stark contrast to research on the role of intent in children's punishment in other domains. Research on children's decisions to punish an agent who has caused physical harm or property damage has found that children begin to take intent into account at 5 to 7 years of age (Baird & Astington, 2004; Cushman et al., 2013; Helwig et al., 2001; Nobes et al., 2016; Nobes, Panagiotaki, & Pawson, 2009; Shultz, Wright, & Schleifer, 1986; Zelazo et al., 1996).

Thus, these results present a puzzle: Why are punishment judgments in property damage or harm cases sensitive to intentionality at 6 years of age, whereas decisions in the ultimatum game show intent sensitivity only around 10 or 12 years? We propose that processing the divider's intent, as it has been implemented in prior versions of the ultimatum game, may be too taxing for young children. In these studies, children must *infer* the agent's intent by comparing the chosen allocation against another available but unchosen alternative (e.g., Sutter, 2007). In contrast, in moral judgment tasks, the actor's intent is *explicitly* referenced (e.g., "He wanted to . . ."). Therefore, we may find that younger children are able to incorporate intent when punishing unfair allocations as long as the divider's intent is explicitly stated and need not be inferred. Consistent with this hypothesis, children as young as 6 years take intent into account when making moral evaluations of free riders, reporting that fictional characters who intentionally free ride are morally worse and less likable than those who do so accidentally (Yang et al., 2018).

Theoretical debates on the role of mutual respect and inequity aversion in fairness

Recently, Engelmann and Tomasello (2019) proposed a theory that views mutual respect as the driver of fairness concerns. Specifically, this theory argues that children's preference for fairness arises not merely from aversion to inequity but also out of a desire to be treated with respect by other social agents. From this perspective, intent should matter especially strongly to children who have been treated unfairly. If a desire to feel respected is the primary underlying motivator of punishment in response to unfairness, the children in our study should punish only in response to intended unfair-

ness and not when it occurs by chance. Moreover, children should be particularly concerned about whether or not the allocation was intended in the second-party condition (and less concerned about intentions in the third-party condition). According to [Engelmann and Tomasello's \(2019\)](#) theory, as children develop, they care first about whether or not they personally are treated with respect and care later about whether or not others are treated with respect as well.

Alternatively, children may also be motivated to punish by an aversion to the inequity itself rather than concerns about respect in particular ([McAuliffe, Warneken, & Blake, 2019](#)). If this is the case, the importance of intent when punishing in response to fairness norm violations may be overridden by an aversion to inequity, whether intended or not, and a desire to create equity whenever possible. Moreover, this view predicts no relationship between intent and punishment perspective given that children are focusing only on the outcome of the allocation in both second- and third-party contexts not how the allocation was created. Under this hypothesis, punishment serves not necessarily as a reaction to being disrespected, or even to deter future norm violations, but rather to create a more equitable allocation of resources.

The current study allowed us to investigate predictions of these competing hypotheses by testing the role of intent in 4.5- to 7.5-year-old children's punishment behavior in response to fairness norm violations when the divider's intent was directly stated. The children in our study were randomly assigned to one of two conditions. In the Intent condition, children were told that they would be playing with another child who decided how candy would be split up between herself or himself and the participant or a third child. By contrast, in the Windfall condition, children were told that the candy would be split up by randomly drawing a card out of a stack that specified how the candy should be divided. Prior research has indicated that young children likely do understand that windfall allocations, like the ones described here, occur because of chance rather than because of an agent's decision ([Shaw & Olson, 2014](#)).

As described above, children in both the Intent and Windfall conditions played both the second- and third-party punishment games. In this way, we were able to systematically evaluate the effect of intent on both second- and third-party punishment behavior and examine whether or not there was an interaction between a perpetrator's intent and the child's punishment perspective.

Method

Children were invited to play both second- and third-party punishment games with absent peer children. Participants were led to believe that these absent peers were real children who had played their part of the game before and would be returning the next day to collect their reward. Comprehension check questions revealed that all participants did believe that their partners were real children (see online [supplementary material](#) for details on all comprehension and manipulation check questions used in this study).

In the Intent condition, children were told that one of the absent children (who we call the divider here) split up some candy (Skittles) between herself (assuming a female divider hereafter) and the recipient—either the participant (in the second-party punishment game) or another absent child (in the third-party punishment game). In the Windfall condition, children were told that the candy was divided by chance (see details below). In both conditions, after the candy was divided, the participant could choose whether or not to pay one piece of candy to punish the divider. If the participant chose to punish the divider, the divider would lose her entire allocation of candy, but the recipient would keep her (assuming a female recipient hereafter) allocation of candy. Critically, this is different from many similar studies in which “punishment” constitutes a complete rejection of the resource division, where both the divider and recipient lose their allocation of the resources (e.g., [McAuliffe et al., 2015](#)). In the current study, we used genuine punishment, where only the divider's allocation was affected. If the participant chose not to punish, both the divider and recipient would receive all their Skittles as allocated.

All measures, manipulations, and exclusions used in this study are described in the sections below.

Participants

We tested 32 4.5- to 5.5-year-olds ($M = 59.96$ months, range = 54.2–65.9; 15 girls) and 33 6.5- to 7.5-year-olds ($M = 83.7$ months, range = 79–89.8; 16 girls). Participants were recruited from a database of children from mixed socioeconomic backgrounds in a major U.S. city.

Sample sizes were selected to reflect the sample sizes used in similar prior research (e.g., McAuliffe et al., 2015). Sample sizes were established a priori, and data collection was stopped after reaching that point. We conducted a sensitivity power analysis to determine the effect sizes that our study would have 80% power to detect given our design and sample size. Using the *simr* package (Green & MacLeod, 2016) in R, we ran a series of 1000-sample bootstrap tests in which we manipulated the desired effect size until we found that our model had $\sim 80\%$ power to detect it. This analysis found that, given our sample size, we had 80% power to detect a small to medium effect (a log odds ratio of ~ 1 or an odds ratio of ~ 2.7) for our three main effects (effects of fairness, punishment perspective, and intent) and three of our primary two-way interactions (age by intent, age by punishment perspective, and age by fairness) (see S1 Table 1 in the [supplementary material](#) for full sensitivity analysis results; see also Chen, Cohen, & Chen, 2010).³

Design

Participants were randomly assigned to either the Intent or Windfall condition (between participants) and were presented with a total of 10 trials, half played in the context of a second-party punishment game and half played in the context of a third-party punishment game. The order of the second- and third-party punishment games was counterbalanced across participants, so that approximately half of the participants played the second-party punishment game first and half played the third-party punishment game first. On each trial, participants decided whether or not to punish a divider based on a specific distribution of candy. Distributions were either unequal (5–1: 5 for divider and 1 for recipient) or equal (3–3). In each game (consisting of 5 trials), participants saw 3 unequal and 2 equal distributions presented in pseudorandom order, such that no more than 2 trials of the same allocation type were presented consecutively. This 3:2 ratio of unequal to equal distributions was selected in order to maximize the number of unequal trials while keeping the total number of trials as low as possible.

Procedure

Children first played a game designed to orient them to the Intent and Windfall manipulations (see [supplementary material](#) for details). Participants did not begin the main experiment until they could correctly answer a series of questions to verify that they were able to distinguish between the Intent and Windfall manipulations and that they understood the difference between an intended outcome and one that happened by chance.

Before the main test, children were given 25 Skittles in what we called their Skittle bags. Participants were then given the instructions for either the second- or third-party punishment game. For both games, participants were told that they would be playing with other children who were of their age and gender. In the second-party game, participants played the role of the recipient and played with one other child who was the divider. In the third-party game, participants observed two other children playing (one who was the divider and the other who was the recipient). Participants were told that all the other children had played their parts of the game “yesterday” because they were not able to come to play today. The experimenter produced Skittle bags with names and faces on them, ostensibly drawn by the absent children, and explained to participants that the other children were returning later to get their Skittle bags.

³ An odds ratio of 1.5 is considered a small effect (equivalent to Cohen's $d = .20$), an odds ratio of 3.5 is a medium effect (equivalent to Cohen's $d = .50$), and an odds ratio of 9 is a large effect (equivalent to Cohen's $d = .80$) (Chen, Cohen, & Chen, 2010).

Participants were informed that because the other children could not be present today, Skittle cards would be used to divide up the Skittles. The face of each Skittle card was divided in half, one side for the divider and the other side for the recipient. For unfair allocations, 5 Skittles were drawn on the divider's half and 1 Skittle was drawn on the recipient's half. For fair allocations, 3 Skittles were drawn on each half of the card. Five cards were used for each game, three with unfair (5–1) allocations and two with fair (3–3) allocations.

In the Windfall condition, the experimenter stated that there were 6 Skittles that needed to be split up between the recipient and the divider and that the experimenter was going to use the Skittle cards to do so. Children watched the experimenter mix up the cards and place them in a card box. The card box had been preloaded with a set of identical cards that were in a predetermined pseudorandom order. The experimenter said that the cards would be drawn from the box one at a time to indicate how the Skittles would be divided for that trial. The cards were loaded into the box in such a way that it appeared that only a single set of cards was being used and that the cards participants had seen being mixed up were the same cards being drawn for each trial of the game.

In the Intent condition, the experimenter stated that it was the divider's job to split up 6 Skittles between herself and the recipient, however she wanted. Furthermore, the divider had drawn how she wanted to split up the Skittles on the cards. Once it was clear that the divider had drawn her decisions on the cards, the cards were placed in the card box as described above. Importantly, on every trial in the Intent condition the Skittles were described as being split up by the divider based on how she wanted the Skittles divided (e.g., "Let's see how Sarah wanted to split up the Skittles," "Sarah wanted to keep 5 Skittles for herself and give 1 Skittle to you"), whereas in the Windfall condition the Skittles were described as being split up based on the cards with no input from the divider whatsoever (e.g., "This card says that 5 Skittles should go to Sarah and 1 Skittle should go to you").

After the cards were put in the card box, participants were introduced to a red box and a green box, which were to be used to indicate their decisions in the game. On each trial, after learning how the Skittles were to be allocated, participants were told that they had the option of doing one of two things. If they took a Skittle out of their Skittle bag and put it in the green box, the Skittles that had been allocated for the participants would be put in their Skittle bags to take home. The Skittles that had been allocated for the divider would be put in the divider's Skittle bag. Then, at the end of the game, all the Skittles in the green box would be put back in the participants' Skittle bags to take home. Therefore, choosing not to punish still required that children make an active decision by placing a Skittle in the green box, but the Skittles were ultimately returned to them. If children took a Skittle out of their Skittle bag and put it in the red box, the Skittles that had been allocated for them would still be put in their Skittle bags to take home. However, the Skittles that had been allocated for the divider would be put in the red box. Children were told that all the Skittles in the red box would be thrown away at the end of the game. Therefore, choosing to punish resulted in the participants' loss of the Skittle they used to indicate their choice and the divider's loss of her Skittle allocation. However, even when choosing to punish, participants kept their allocation of Skittles (see Fig. 1).

Children received instructions twice, once before playing the second-party game and once before playing the third-party game. After each set of instructions, they were asked a number of comprehension check questions. Children were asked these comprehension check questions twice, once after receiving the instructions for the second-party game and once after receiving the instructions for the third-party game. Participants were excluded from analyses if they incorrectly answered whether the Skittles were divided intentionally or by chance both times they were asked. They were also excluded if they answered fewer than two thirds of the other comprehension check questions correctly. In total, 3 participants were excluded for failing to meet these standards (2 younger children and 1 older child), and 1 participant (a younger child) was excluded for failing to complete the entire study (see SI Table 2 in the [supplementary material](#) for a question-by-question breakdown of correct response rates in each age group).

Each trial began with the experimenter drawing a card out of the box and then dividing the Skittles appropriately as the participants watched. If it was the first trial of the game, children were then asked to verify whether the allocation had been made intentionally by the divider or by chance. Once again, only children who responded correctly to this question were included in our data analysis. Participants then made their decisions by taking a Skittle from their Skittle bag and placing it in either

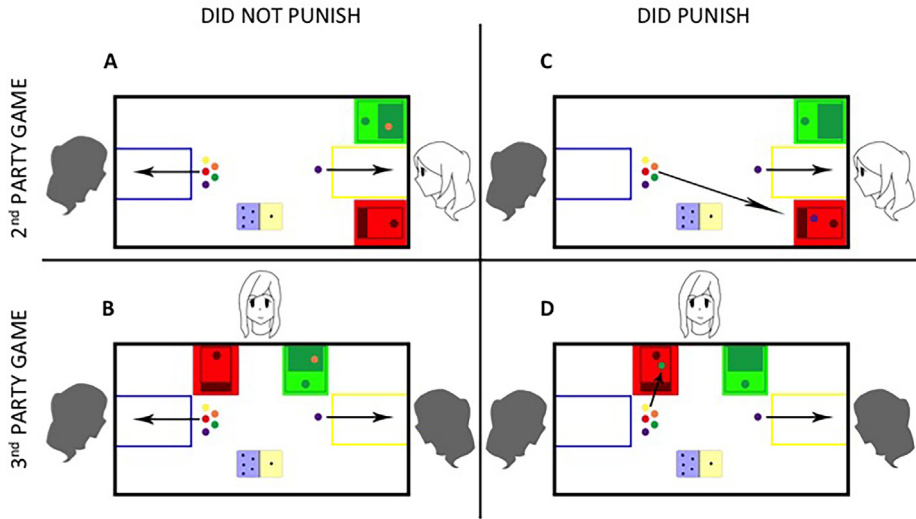


Fig. 1. The experimental environment. Participants (shown here as the child with a visible face) play with either one other absent child (A and C) or two other absent children (B and D), shown here as gray silhouettes. Allocations are indicated by the blue and yellow Skittle cards, with Skittles on the blue side going to the divider and those on the yellow side going to the recipient. Participants make their decisions by putting a Skittle in either the green box to accept the allocation (A and B) or the red box to punish the divider (C and D). If the allocation is accepted, both the divider and recipient keep their parts of the allocation. If the participant chooses to punish, the recipient keeps her share of the allocation, but the divider's share goes in the red box. At the end of the game, the participant keeps all the Skittles in the green box and all the Skittles in the red box are thrown away. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

the red or green box. After the decision was made, the experimenter appropriately distributed the Skittles. The study ended once each child had completed 5 trials of both games.

To directly probe children's explanations for their decisions and gain potential insight into their motivations for these decisions, after children had completed both games, they were asked why they had decided to put Skittles in the red box and why they had decided to put Skittles in the green box (more details on children's responses can be found in the [supplementary material](#)). Finally, to test whether participants believed that they were playing with other real children, as they were leaving a second experimenter reentered the room and asked whether the Skittle bags (belonging to the fictional other children) should be saved for kids who were going to come back later or if they were really just for pretend kids. Every child responded that the Skittle bags should be saved for the kids who were going to come back later to get them. Participants were then escorted out of the study room and given a prize for their participation.

Data coding and analysis

All sessions were videotaped. Children's decisions were live coded by an observer from an adjacent room over video feed. Our main variable of interest was whether or not children decided to punish the divider. A randomly selected subset of 12 testing sessions (18.5%) was later recoded by an independent video coder who was blind to the hypothesis. Correspondence between the live coding of children's decisions and the recoding from video was 100%, therefore, no additional sessions were recoded.

Data were analyzed using R statistical software. Binary punishment responses (0 = did not punish, 1 = punished) were analyzed using mixed-effects logistic regression implemented using the *lme4* package (Bates, Maechler, Bolker, & Walker, 2015). We included fixed-effects categorical predictors for age (4.5–5.5 years or 6.5–7.5 years), condition (Intent or Windfall), punishment perspective

(second-party or third-party punishment), and fairness (fair or unfair) as well as all possible interactions between these predictors. This model also included a random intercept for each participant and random slopes and their interaction for the within-participant factors (fairness and punishment perspective). We used a hypothesis-driven model comparison approach, with statistical significance evaluated using the *pbrtest* package (Halekoh & Højsgaard, 2014). By using parametric bootstrapping to perform model comparisons, we did not rely on as many assumptions regarding the underlying distribution as traditional methods do for assessing significance.⁴ Specifically, we compared this full model with a reduced model that did not include the highest-level interaction term (i.e., the four-way interaction among age, condition, punishment perspective, and fairness). If this interaction term was not significant, we compared this reduced model to one that did not include the next highest-level interaction term(s) (i.e., all three-way interactions). Once interactions at a particular level were significant, we probed these interactions directly. In this way, we minimized the number of tests performed and thus minimized the likelihood of false-positive results. In addition to this set of primary analyses, we included several additional exploratory analyses (e.g., first-trial behaviors) that are included in the [supplementary material](#).

Results

We first examined our data for the presence, in turn, of higher-order interactions. We did not find any four-, three-, or two-way interactions (all 5000-sample bootstrap likelihood ratio test [LRT], $\chi^2 < 3.44$, $\beta = -1.36$, $p > .32$). However, a model including only main effects did perform better than a null model without any main effects (5000-sample bootstrap LRT, $\chi^2(1) = 17.51$, $p < .005$). To determine which predictors were contributing significantly, we compared a model containing all main effects with a model dropping each of the predictors in turn. We found that only fairness and punishment perspective improved model fit (fairness: 5000-sample LRT, $\chi^2(1) = 11.04$, $p < .005$; punishment perspective: LRT, $\chi^2(1) = 9.30$, $p < .005$); therefore, our final model contained only these two predictors. We now discuss each of these effects in detail.

First, as predicted, we found a significant main effect of fairness on punishment ($\beta = 1.36$, $SE = 0.47$, $p < .005$). Across age group and condition, children were more likely to punish the divider when the Skittles were allocated unfairly than when they were divided fairly.

We also found a significant main effect of punishment perspective on punishment ($\beta = -1.13$, $SE = 0.39$, $p < .005$). Across age group, fairness, and condition, children were more likely to punish a divider when they were direct recipients of an allocation than when they were observers. There were no significant interactions between age and punishment perspective (LRT, $\chi^2(1) = 0.58$, $\beta = -0.86$, $p = .46$) or among age, punishment perspective, and fairness (LRT, $\chi^2(1) = 0.74$, $\beta = 1.50$, $p = .12$).

Given that this project built off of previous work by McAuliffe et al. (2015) on the development of third-party punishment, we did a secondary analysis using their analytic strategy. In particular, McAuliffe and colleagues operationalized punishment as significantly more punishment in response to unfair offers than in response to fair offers. Using this methodology, we found that in the second-party condition, older children punished unfair offers significantly more than they punished fair offers and younger children demonstrated a similar pattern of responses (although the difference was not quite significant for younger children [older: LRT, $\chi^2(1) = 5.28$, $p = .03$, $\beta = 1.19$, $SE = .60$; younger: LRT, $\chi^2(1) = 3.49$, $p = .097$, $\beta = 1.11$, $SE = 0.06$]). However, in the third-party condition, only older children showed this preference for punishing in response to unfair offers relative to fair offers [older: LRT, $\chi^2(1) = 6.62$, $p = .01$, $\beta = 5.96$, $SE = 2.53$; younger: LRT, $\chi^2(1) = 0.95$, $p = .43$, $\beta = 0.99$, $SE = 1.25$] (see Fig. 2).

We found no main effects of age or condition (our Intent vs. Windfall manipulation on punishment decisions) [age: LRT, $\chi^2(1) = 0.60$, $\beta = 0.21$, $p = .49$; condition: LRT, $\chi^2(1) = 0.02$, $\beta = 0.41$, $p = .90$]. We also found no interactions between our Intent/Windfall manipulation and any other variable (see Fig. 3 and SI Table 4 in the [supplementary material](#)). These results are consistent with the hypothesis

⁴ Note that this also means that the significance value for a particular comparison might not be the same as would be expected from a standard likelihood ratio test. See the documentation of the *pbrtest* package for more details (Halekoh & Højsgaard, 2014).

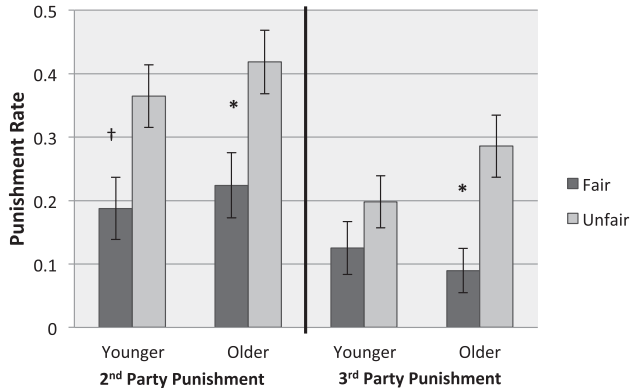


Fig. 2. Punishment rates in response to fair versus unfair offers for 4.5- to 5.5-year-olds and 6.5- to 7.5-year-olds in both the second- and third-party punishment games. Error bars represent 1 standard error. † $p < .10$; * $p < .05$.

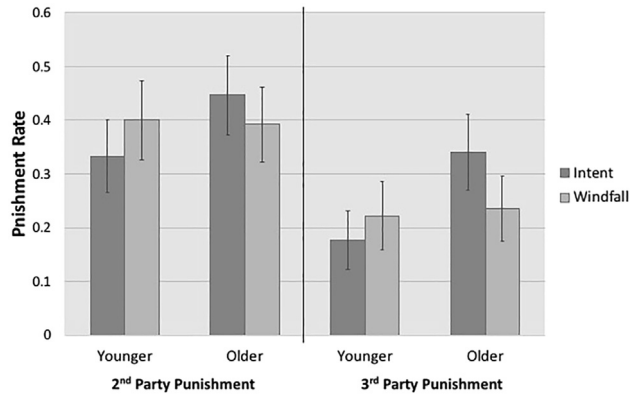


Fig. 3. Punishment rates in response to unfair offers only in the Intent versus Windfall conditions. Responses are divided by age group (4.5–5.5 years or 6.5–7.5 years) and second- or third-party punishment game. Error bars represent 1 standard error. All comparisons are not significant.

that there is no interaction between a norm violator's intent and second- versus third-party punishment. Rather, children's responses to fairness norm violations appear to be entirely focused on outcome. Specifically, children's punishment behavior depends on (a) whether or not the outcome was fair or unfair and (b) whether or not the children were the victims.

Finally, we found no differences between male and female participants and no order effects (see [supplementary material](#) for details). In other words, children punished at the same rate regardless of which game they played first.

Discussion

This is the first study to directly compare the development of costly second- and third-party punishment in response to unfairness by requiring young children to make costly punishment decisions in both contexts. We found that both younger children's (4.5–5.5 years) and older children's (6.5–7.5 years) punishment behavior was sensitive to the fairness of an allocation and whether or not the children were the direct recipients of the allocation. We also found that whether allocations were

made intentionally or by chance did not affect children's punishment behavior in our study. These findings are consistent with the hypothesis that children younger than 8 years are primarily focused on the outcome of an allocation rather than the intention behind it when punishing in response to fairness norm violations. Moreover, our results highlight that being a victim of unfairness has a major impact on punishment behavior; children will stand up for others, but they are much more likely to stand up for themselves.

Effect of fairness on punishment

As expected, we found that across age group, punishment perspective, and condition (Intent or Windfall), children punished significantly more often in response to unfair allocations than in response to fair allocations. These findings confirm that children were sensitive to inequity even in the third-party context.

Effect of second- and third-party perspectives on punishment

We found that across age group, fairness, and condition (Intent or Windfall), children punished significantly more often in the second-party context than in the third-party context; children were much more likely to punish when they were direct victims of unfairness than when they were merely observing another child being treated unfairly. We did not find a significant interaction among age, fairness of the allocation, and punishment perspective. However, a post hoc power analysis revealed that we had insufficient power to detect a three-way interaction given our sample size and observed effect sizes. Despite the lack of an interaction, closer examination of the data revealed that whereas both younger and older children tended to punish in response to unfair offers more often than in response to fair offers in the second-party punishment game, only older children punished in response to unfair offers significantly more often than in response to fair offers in the third-party punishment game.

Our findings provide new insights into the emergence of children's willingness to pay a cost to intervene against inequality. Whereas 6- and 7-year-olds intervene reliably in response to unfairness in both second- and third-party contexts, 4- and 5-year-olds are only starting to engage in these behaviors. This is consistent with findings that children first begin paying a cost to prevent inequality as a second party at around 4 or 5 years of age (e.g. Bereby-Meyer & Fiks, 2013; Blake & McAuliffe, 2011). Moreover, work on costly third-party punishment showed that children from a similar population reliably intervene from 6 years of age onward but show only a trend at 5 years (Jordan et al., 2014; McAuliffe et al., 2015). Critically, our work expands on this previous research by showing that this delayed emergence of third-party punishment relative to second-party punishment persists even when the intervention is genuine punishment. In prior research, "punishment" constituted a complete rejection of the resource division, where both the divider and the recipient lose their allocation of the resources (Jordan et al., 2014; McAuliffe et al., 2015). Typically, however, punishment is most often directed only toward the transgressor and the cost of punishment is incurred only by the punisher, not the victim (e.g., someone who is robbed does not need to pay the cost of jailing the perpetrator). Thus, prior evidence of later-emerging third-party punishment may have reflected an unwillingness to impose a cost on an innocent individual rather than an unwillingness to punish bad behavior. Our findings indicate that even with a more targeted infliction of cost, children seem unwilling or unable to engage in third-party punishment until 1 or 2 years after they begin engaging in second-party punishment.

Our results fit well in the broader context of how children's internalization of fairness norms develops as children age. Past research has found that young children are particularly concerned with their own gains and are sensitive primarily to their own inequity rather than to fairness in general (Benenson, Pascoe, & Radmore, 2007; LoBue et al., 2011; Sheskin, Bloom, & Wynn, 2014). However, as children age, they increasingly value fairness norms until 8 years, when they will reject unfair offer allocations even when they are not the beneficiaries of inequity (Blake & McAuliffe, 2011).

The singular focus of 4- and 5-year-olds on their own inequity may be driving the effects we see in the current study. By contrast, our older children are more likely to engage in costly third-party pun-

ishment that requires greater perspective taking or a desire to enforce fairness norms more generally (McAuliffe et al., 2017). These older children are concerned less with who the victim is and more with whether or not fairness norms were violated at all. Consistent with these conclusions, a recent study found that 6-year-olds, but not 4-year-olds, will pay a cost to watch an antisocial agent be punished, suggesting that younger children prioritize their own welfare, whereas older children more strongly value norm enforcement (Mendes, Steinbeis, Bueno-Guerra, Call, & Singer, 2018).

This might reflect a broader developmental trajectory toward increasingly more principle-based decision making. Specifically, Jordan et al. (2014) found that 6-year-olds were more likely to engage in costly third-party punishment when the victim was an ingroup member than when the victim was an outgroup member, but 8-year-olds defended ingroup and outgroup victims alike. This would suggest that young children may be driven to punish by personal motives, but as they age they shift their focus to norm enforcement in a more impersonal and more principled way. Obviously, for children as for adults, there remains a tension between more self-focused motives and what is expected by fairness norms, and while children become relatively more fairness-minded, egoistic and parochial tendencies never completely disappear.

Effect of intent on punishment

In the current study, we made the divider's intent explicit, telling children directly whether the Skittles were divided by another child intentionally or whether the allocations were a result of chance. Despite the directness of the intent manipulation, we found that young children still do not seem to be consistently taking intent into account when punishing in response to fairness norm violations. Our results are commensurate with previous research finding that children do not consider intent when deciding whether or not to punish an agent who violates fairness norms until around 10 or 12 years of age (Bueno-Guerra et al., 2016; Gummerum & Chu, 2014; Güroğlu et al., 2009; Sutter, 2007). However, a separate body of research has found that children between 3 and 7 years of age begin taking intent into account when making punishment decisions in response to physical harm or property damage (Baird & Astington, 2004; Cushman et al., 2013; Helwig et al., 2001; Nobes et al., 2009, 2016; Shultz et al., 1986; Zelazo et al., 1996). What, then, explains the discrepancy between findings that children under 10 years of age do not care about intent when punishing in response to unfairness, as shown here and by others (Bueno-Guerra et al., 2016; Gummerum & Chu, 2014; Güroğlu et al., 2009; Sutter, 2007), and the literature suggesting that they do consider intent when deciding to punish in response to physical harm?

One possibility is that children in the current study did not comprehend the difference between the Intent and Windfall conditions. However, several features of our procedure showed that participants understood the distinction between the two conditions. All participants were trained extensively on the Intent manipulation prior to beginning the task. Children moved to the test phase only once they demonstrated that they understood the difference between an intentional decision and one that is made by chance. Finally, children needed to correctly answer a comprehension check question about whether the Skittles were divided intentionally or by chance after the first trial of each game. Any child who did not answer this question correctly was excluded from analysis.

Alternatively, it may be that young children punish in response to fairness norm violations for different reasons than they do in response to personal harm. Recently, Engelmann and Tomasello (2019) proposed that children punish unfairness because they feel disrespected and not because they disapprove of inequity per se. Under this hypothesis, children should be especially willing to punish when unfairness is intended rather than a chance outcome. However, in our study children were willing to pay a cost to punish in both the Intent and Windfall conditions. Instead, our results are consistent with the alternative—that children's punishment behavior is in part about trying to rectify disadvantageous inequity rather than in response to feeling disrespected (McAuliffe et al., 2019). Continuing to explore the differences between the motivations to punish in response to harm versus unfairness remains an important area for future research.

Acknowledgments

This work was supported by the Barbara Ditmars Fund at Harvard University (to R.M.B.) and by a National Science Foundation CAREER award (to F.W.). We thank Redaate Wolle and Cassandra Favart for their administrative and research support and thank Stefen Beeler for constructing experimental props. The Fig. 1 design credit goes to Devon Gunter.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jecp.2020.104909>.

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